Detailed Contents

1. **Analog and Digital Signals** **1–4**
   Definitions • Difference between analog and digital signals • Advantage of Digital signal over analog signal.
2. **Number Systems** **5–18**
   Definitions • Radix representation of numbers • Conversion of one number system to the other (Decimal, Binary, Octal and Hexadecimal) • Fixed point representation • Floating point representation.
3. **Logic Gates** **19–34**
   Definitions • Basic Gates-NOT, OR, AND • Universal gates-NOR, NAND • X-OR gate • AND-OR-Invert gates • De-Morgan's laws • Logic circuit for Boolean Expression and vice-versa • Positive logic and negative logic.
4. **Combinational Logic Circuits and Simplification** **35–54**
   Boolean laws and theorems • Consensus theorem • Karnaugh Map • Don't care condition • Minterm and Maxterm • Simplification using Boolean laws and theorems • Simplification using Karnaugh map for 3, 4 and 5 variables—SOP and POS • Simplification using tabular method • Simplification using Consensus theorem • Hazard and Hazard cover.
5. **Binary Arithmetic** **55–74**
   Binary addition, subtraction, multiplication, division • I's complement and 2's complement • Half Adder • Full Adder • Half Subtractor • Full Subtractor • Adder-Subtractor.

6. Codes and Parity **75–90**

Weighted code-BCD code • Non weighted codes-Excess 3 code, Gray code • Alpha numeric codes-ASCII, EBCDIC • Error detecting codes-Odd parity, Even parity (Single and Double Error) • Error correcting or Hamming code • Checksum code • CRC code.

7. Multiplexer, Demultiplexer, Encoder, Decoder **91–120**

Definitions • Multiplexers • Demultiplexers • Implementation of logic function using multiplexer • Decoder, BCD to decimal decoder • Seven segment indicator • Encoder, decimal to BCD encoder.

8. Different Types of Displays **121–128**

LED • LCD • Advantage and disadvantage • Difference between LED ad LCD • Alpha numeric display-discrete method, bar-matrix method and dot-matrix method.

9. Digital Logic Families **129–152**

Bipolar Logic Family • Saturated-RTL, IIL, DTL, HTL and TTL • Non-Saturated Schottky TTL, ECL • Unipolar logic family-PMOS, NMOS and CMOS • SSI, MS LSI, VLSI • Characteristics of digital ICs.

10. Flip Flops **153–174**

Definitions • RS Flip Flop • D Flip Flop • JK Flip Flop • T Flip Flop • Conversion of one Flip Flop to other • Finite state machine • Moore and Mealy model • State graph and state tables.

11. Shift Registers **175–190**

Definitions • Type of registers • Serial In-Serial Out • Serial In-Parallel Out • Parallel In-Serial Out • Parallel In-Parallel Out • Universal shift register • Ring counter.

12. Counters **191–214**

Introduction • Types of counter • Asynchronous counter • Synchronous counter • Difference between Asynchronous and Synchronous counter • Presettable counter • Design of counters for different modules • Design of binary counters of random sequence using flip flop.

13. Semiconductor Memories **215–226**

Definitions • Description RAM and ROM • Semiconductor memory organization • Classification • Expansion of word length and word capacity • Difference between Static RAM and Dynamic RAM.

14. D/A Conversion and A/D Conversion **227–250**

D/A Converter • Binary equivalent rate • Binary Ladder • A/D Converter Simultaneous, Counter Type, Continuous A/D converter, successive approximation method, single slope and dual slope A/D converter.